

Section I

10 marks

Attempt Questions 1 – 10

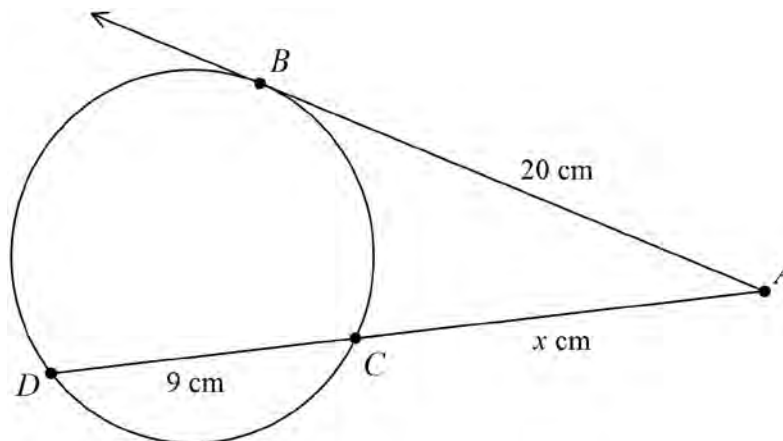
Use the multiple choice answer sheet located at the back of the paper.

Allow about 15 minutes for this section

1. If A and B are the points $(2, -1)$ and $(-2, -4)$ respectively, the coordinates of the point M which divides AB externally in the ratio $1:3$ is,

(A) $M\left(4, \frac{1}{2}\right)$ (B) $M\left(-2, -\frac{11}{2}\right)$ (C) $M\left(-2, -\frac{7}{4}\right)$ (D) $M\left(2, -\frac{13}{4}\right)$

2.



$AB = 20$ cm, $CD = 9$ cm and $AC = x$ cm. Find the value of x .

- (A) 5 (B) 9 (C) 11 (D) 16

3. The derivative of $3\sin^{-1}\frac{x}{2}$ is,

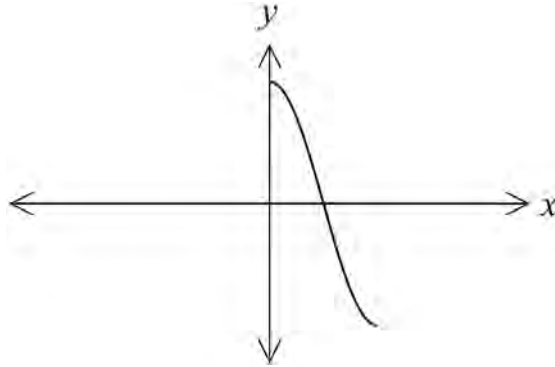
(A) $\frac{3}{\sqrt{4-x^2}}$ (B) $\frac{3}{\sqrt{1-4x^2}}$
(C) $\frac{3}{\sqrt{\frac{1}{4}-x^2}}$ (D) $\frac{3}{\sqrt{1-\frac{x^2}{4}}}$

4. The exact value of $\cos 105^\circ$ is,

(A) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ (B) $\frac{1-\sqrt{3}}{2\sqrt{2}}$ (C) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ (D) $\frac{1+\sqrt{3}}{2\sqrt{2}}$

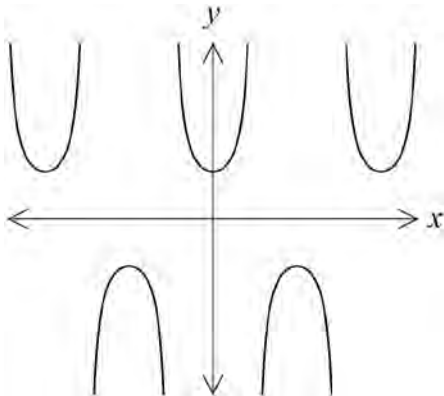
5. When the polynomial $P(x)$ is divided by $x^2 - 4$, the remainder is $2x - 3$. What is the remainder when $P(x)$ is divided by $x + 2$?
- (A) -7 (B) 1 (C) 5 (D) 7

6. The diagram shows the graph $y = f(x)$.

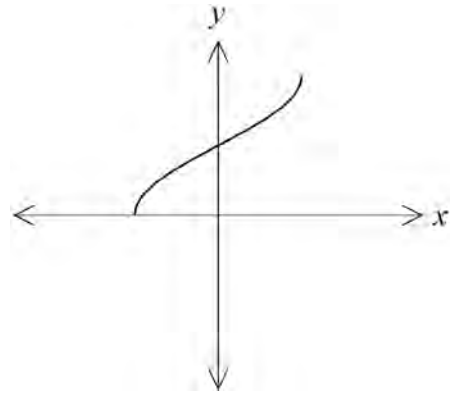


Which diagram shows the graph of $y = f^{-1}(x)$

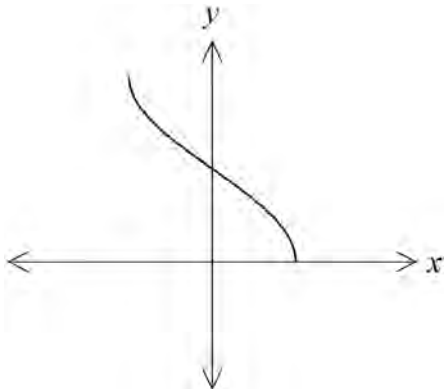
(A)



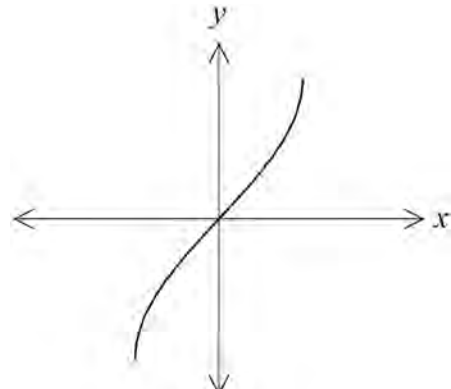
(B)



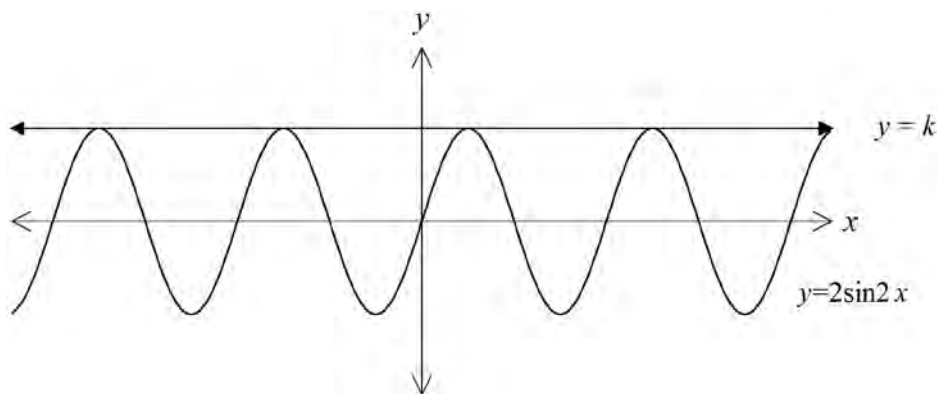
(C)



(D)



7. A curve is defined by the parameters $x = 2p + \frac{2}{p}$, $y = p^2 + \frac{1}{p^2}$. Which of the following represents this curve in Cartesian form?
- (A) $y = \frac{x^2}{4} - 2$ (B) $y = \frac{x^2}{2}$ (C) $x + y^2 = 2$ (D) $y = x^2 - 2$
8. Which of the following is an expression for $1 + \sec x$ in terms of $t = \tan \frac{x}{2}$?
- (A) $\frac{2}{1+t^2}$ (B) $\frac{2}{1-t^2}$ (C) $\frac{2t^2}{1+t^2}$ (D) $\frac{2t^2}{1-t^2}$
9. Which integral is obtained when the substitution $u = x + 2$ is applied to $\int \frac{x}{3} \sqrt{x+2} \, dx$?
- (A) $\frac{1}{3} \int (u^{\frac{1}{2}} + 2u^{\frac{1}{3}}) du$ (B) $\frac{1}{3} \int (u^{\frac{1}{2}} - 2u^{\frac{3}{2}}) du$
- (C) $\frac{1}{3} \int (u^{\frac{3}{2}} + 2u^{\frac{1}{2}}) du$ (D) $\frac{1}{3} \int (u^{\frac{3}{2}} - 2u^{\frac{1}{2}}) du$
10. Part of the graph of $y = 2 \sin 2x$ is drawn below.



The horizontal line $y = k$ is also drawn to touch $y = 2 \sin 2x$ as shown above.
 Given that n is an integer, the general solution of the intersection of these two functions is,

- (A) $\frac{n\pi}{2} + (-1)^n \frac{\pi}{4}$ (B) $2n\pi \pm \frac{\pi}{4}$ (C) $n\pi + (-1)^n \frac{\pi}{2}$ (D) $\frac{n\pi}{2} + (-1)^n \frac{\pi}{2}$

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available upon request from the supervising teachers.

In questions 11-14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks) Use a SEPARATE writing booklet.

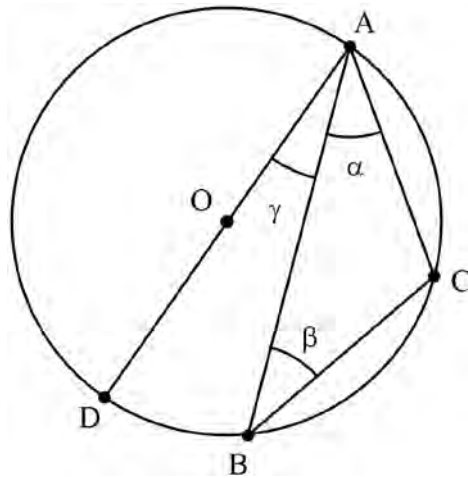
Marks

- (a) Find the value of $\lim_{x \rightarrow 0} \frac{\tan 3x}{2x}$. 1
- (b) Find the gradient of the tangent to the curve $y = \cos(\ln x)$ when $x = 1$. 2
- (c) Find $\int \frac{1}{4 + 9x^2} dx$ 2
- (d) Evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos^2 3x dx$ 3
- (e) Solve $\frac{x^2 + x - 6}{x} \geq 2$. 3
- (f) The curves $y = \sin x$ and $y = \cos x$ intersect at $P\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$. If α is the acute angle between these curves at P , then show that $\tan \alpha = 2\sqrt{2}$ 2

Question 11 continues

(g)

2



The diagram shows points B and C on a circle with centre O and diameter AD , as shown in the diagram.

Let $\angle BAC = \alpha$, $\angle ABC = \beta$ and $\angle OAB = \gamma$.

Copy or trace this diagram into your writing booklet.

Find the value of $\alpha + \beta + \gamma$, giving reasons for your answer.

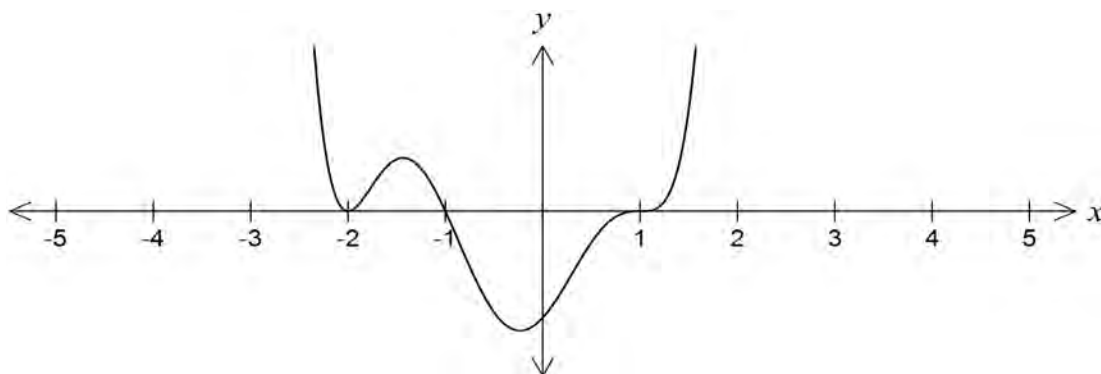
End of Question 11

Question 12 (15 Marks) Use a SEPARATE writing booklet.

Marks

- (a) Write a polynomial equation for the graph $y = P(x)$ shown below.

1



- (b) Prove by mathematical induction, that for $n = 1, 2, 3, \dots$

3

$$1 + (1 + 2) + \dots + (1 + 2 + \dots + 2^{n-1}) = 2^{n+1} - n - 2$$

- (c) The equation $x^3 + x^2 - 4x - k^2 = 0$, where $k > 0$ has only positive roots. If one of the roots is the product of the other two roots.

- (i) Show that $x = k$ is a root of the equation.

2

- (ii) Hence or otherwise find the value of k .

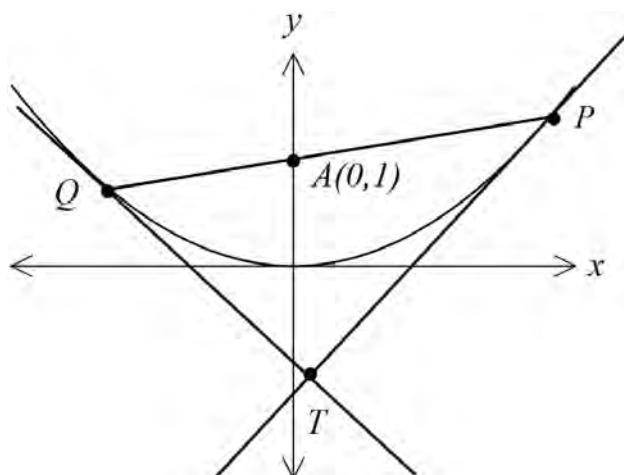
2

- (d) The equation $x^2 - \sqrt{x} - 2 = 0$ has a root near $x = 2$. Use Newton's Method once to obtain a better approximation of x . Answer to 2 decimal places.

2

Question 12 continues

(e)

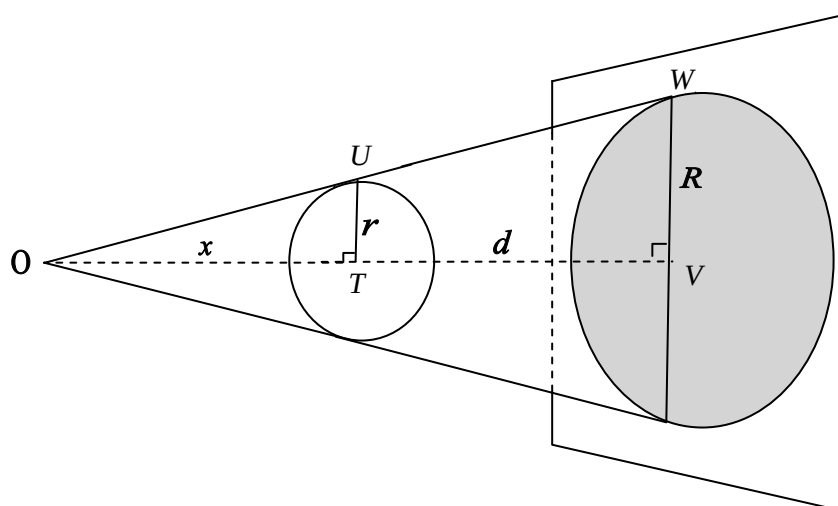


The points $P(4p, 2p^2)$ and $Q(4q, 2q^2)$ lie on the parabola $x^2 = 8y$. The tangents to the parabola at P and Q intersect at T . The chord PQ passes through the point $A(0, 1)$.

- | | | |
|-------|--|----------|
| (i) | Write down the equation of the tangent at P . | 1 |
| (ii) | Hence or otherwise find the coordinates of T . | 1 |
| (iii) | Show that the equation of the chord PQ is given by $2y = (p + q)x - 4pq$ | 1 |
| (iv) | Show that $pq = -\frac{1}{2}$ | 2 |

End of Question 12

(a)



A coin of radius r cm is placed x cm from the light source O , such that its horizontal axis of symmetry passes through O .

A coin is placed d cm away from a screen.

The light source O is moving horizontally towards the coin at a speed of 4cm per second and casts a circular shadow of radius R cm on the screen as shown.

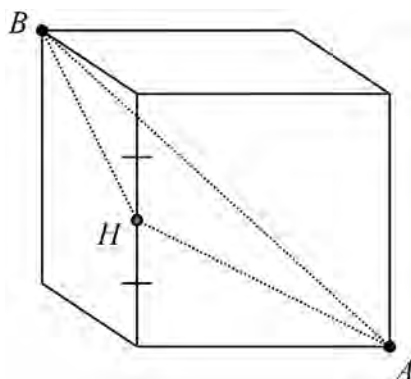
(i) Show that the area of the shadow is $A = \frac{\pi r^2 (x + d)^2}{x^2}$. 2

(ii) Find the rate of increase, $\frac{dA}{dt}$ of the area of the shadow when $x = d$. 2

(b) Prove that $\tan^{-1} \frac{2}{3} + \cos^{-1} \frac{2}{\sqrt{5}} = \tan^{-1} \frac{7}{4}$. 3

Question 13 continues

(c)



The diagram shows a cube with edge length 2 units. H is the midpoint of the edge as shown in the diagram.

Using triangle AHB or otherwise, find the size of $\angle AHB$

2

(d) A particle moves in a straight line and its position at time t seconds is given by,

$$x = 4 + \frac{\sin 4t}{\sqrt{3}} - \cos 4t$$

(i) Express $\frac{\sin 4t}{\sqrt{3}} - \cos 4t$ in the form $R \sin(4t - \alpha)$, where α is in radians.

2

(ii) The particle is undergoing Simple Harmonic Motion, show that the equation for acceleration is,

2

$$\ddot{x} = -16(x - 4)$$

(iii) When does the particle first reach its maximum speed?

2

End of Question 13

Question 14 (15 Marks) Use a SEPARATE writing booklet.

Marks

(a) By expanding both sides of the identity

2

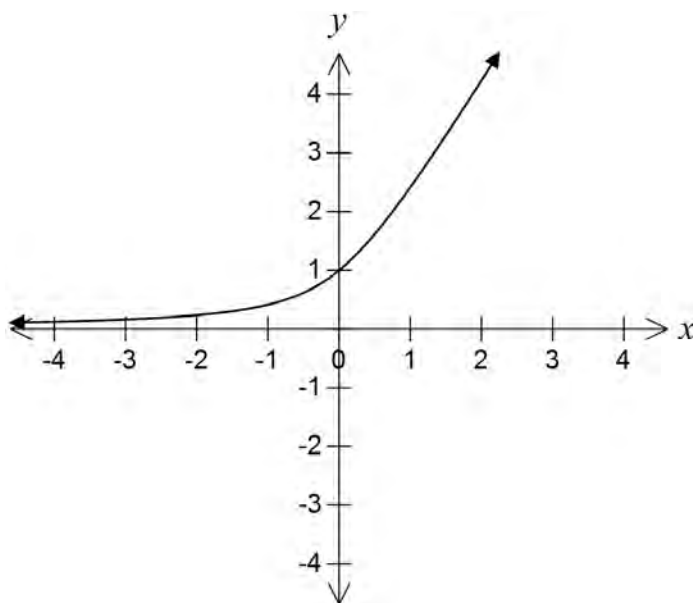
$(1+x)^{n+4} = (1+x)^n (1+x)^4$ prove that

$$\binom{n+4}{r} = \binom{n}{r} + 4\binom{n}{r-1} + 6\binom{n}{r-2} + 4\binom{n}{r-3} + \binom{n}{r-4}, \quad \text{for } r = 4, 5, \dots, n$$

(b) Using the substitution $u = \tan x$, find $\int (\tan^3 x \sec^2 x) dx$

2

(c) Consider the function $f(x) = x + \sqrt{x^2 + 1}$



(i) State the range of $f(x)$

1

(ii) Show that $f'(x) = \frac{f(x)}{\sqrt{x^2 + 1}}$ and hence show that $f'(x) > 0$ for all x in the domain.

2

(iii) State the range of the inverse function of $f(x)$.

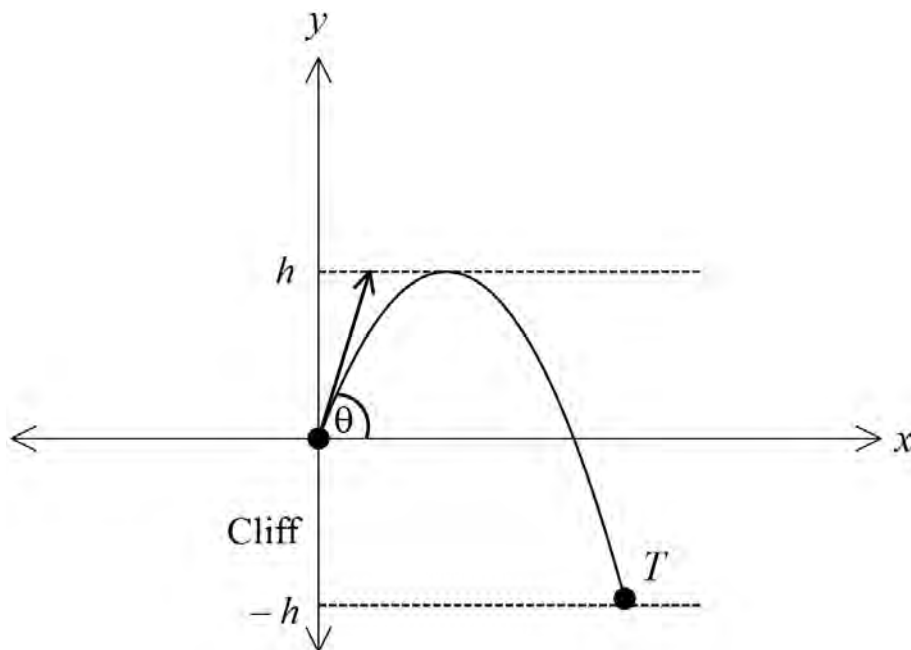
1

(iv) Show that the inverse function is $f^{-1}(x) = \frac{1}{2} \left(x - \frac{1}{x} \right)$.

2

Question 14 continues

(d)



The path of a projectile fired from the top O of a cliff is shown in the diagram above.

Its initial velocity is V m/s at an angle θ to the horizontal plane at O . It rises to a maximum height h metres above O and strikes a target T on the ground h metres below O .

Assuming the usual horizontal and vertical components of displacement in metres at time t seconds, are $x = Vt \cos \theta$ and $y = Vt \sin \theta - \frac{1}{2}gt^2$ respectively.

(i) Prove that $h = \frac{V^2 \sin^2 \theta}{2g}$. 2

(i) Prove that the time taken for the projectile to reach its target is, 2

$$\frac{V \sin \theta (1 + \sqrt{2})}{g} \text{ seconds}$$

(ii) Hence, show that the distance from the base of the cliff to the target is, 1

$$\frac{V^2 (1 + \sqrt{2}) \sin 2\theta}{2g}$$

End of paper

2014 CTHS Mathematics Extension 1 AP4 Solutions

SECTION 1 MULTIPLE CHOICE

- | | Working | Answer |
|----|---|---------------|
| 1. | $(2, -1) \quad (-2, -4)$
$-1:3$
$x = \frac{-1 \times -2 + 3 \times 2}{-1 + 3} \quad y = \frac{-1 \times -4 + 3 \times -1}{-1 + 3}$ $= 4 \quad = \frac{1}{2}$ | A |
| 2. | $x(x+9) = 400$
$x^2 + 9x = 400$
$x^2 + 9x - 400 = 0$
$(x+25)(x-16) = 0$
$x = -25$ or $x = 16$
x is a length
$\therefore x = 16$ | D |
| 3. | $\frac{d}{dx} \left(3 \sin^{-1} \frac{x}{2} \right) = \frac{3}{\sqrt{2^2 - x^2}}$ $= \frac{3}{\sqrt{4 - x^2}}$ | A |
| 4. | $\cos 105^\circ = \cos(60^\circ + 45^\circ)$
$= \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$
$= \frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$
$= \frac{1 - \sqrt{3}}{2\sqrt{2}}$ | B |
| 5. | $P(x) = Q(x)(x^2 - 4) + 2x - 3$
$P(-2) = 0 + 2 \times -2 - 3$
$= -7$ | A |
| 6. | Inverse cos graph | C |

- Working** **Answer**
7. **A**
- $$x = 2p + \frac{2}{p} \quad y = p^2 + \frac{1}{p^2}$$
- $$x = 2\left(p + \frac{1}{p}\right) \dots\dots(1)$$
- $$y = \left(p + \frac{1}{p}\right)^2 - 2p \times \frac{1}{p} \dots\dots(2)$$
- $$= \left(p + \frac{1}{p}\right)^2 - 2$$
- $$= \left(\frac{x}{2}\right)^2 - 2 \dots\dots(1) \text{ in } (2)$$
- $$y = \frac{x^2}{4} - 2$$
8. **B**
- $$\cos \theta = \frac{1-t^2}{1+t^2}$$
- $$1 + \sec \theta = 1 + \frac{1+t^2}{1-t^2}$$
- $$= \frac{1-t^2+1+t^2}{1-t^2}$$
- $$= \frac{2}{1-t^2}$$
9. **D**
- $$\int \frac{x}{3} \sqrt{x+2} \, dx$$
- $$u = x+2$$
- $$x = u-2$$
- $$\frac{du}{dx} = 1$$
- $$du = dx$$
- $$\therefore \int \frac{x}{3} \sqrt{x+2} \, dx = \int \left(\frac{u-2}{3} \sqrt{u} \right) du$$
- $$= \frac{1}{3} \int ((u-2)\sqrt{u}) \, du$$
- $$= \frac{1}{3} \int \left(u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \right) du$$

10.	Working $2 \sin 2\theta = 2$ $\sin 2\theta = 1$ $2\theta = \pi n + (-1) \sin^{-1}(1)$ $\theta = \frac{\pi n}{2} + \frac{(-1)}{2} \times \frac{\pi}{2}$ $\theta = \frac{\pi n}{2} + \frac{(-1)\pi}{4}$	Answer A
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SECTION 2 SHORT ANSWER

11.	Solution	Marks
(a)	$\begin{aligned}\lim_{x \rightarrow 0} \frac{\tan 3x}{2x} &= \frac{1}{2} \lim_{x \rightarrow 0} \frac{\tan 3x}{x} \\ &= \frac{1}{2} \times \frac{3}{3} \lim_{x \rightarrow 0} \frac{\tan 3x}{x} \\ &= \frac{3}{2} \lim_{x \rightarrow 0} \frac{\tan 3x}{3x} \\ &= \frac{3}{2} \times 1 \\ &= \frac{3}{2}\end{aligned}$	1
(b)	$y = \cos(\ln x)$ $\frac{dy}{dx} = -\sin(\ln x) \times \frac{1}{x}$ When $x = 1$ $\begin{aligned}\frac{dy}{dx} &= -\sin(\ln 1) \times \frac{1}{1} \\ &= 0\end{aligned}$ $\therefore$ the gradient of the tangent to the curve $y = \cos(\ln x)$ is 0.	2
(c)	$\begin{aligned}&\int \frac{1}{4 + 9x^2} dx \\ &= \int \frac{1}{9\left(\frac{4}{9} + x^2\right)} dx \\ &= \frac{1}{9} \int \frac{1}{\frac{4}{9} + x^2} dx \\ &= \frac{1}{9} \times \frac{3}{2} \tan^{-1}\left(\frac{3}{2} \times x\right) + C \\ &= \frac{1}{6} \tan^{-1}\left(\frac{3x}{2}\right) + C\end{aligned}$	2

11.
(d)

Solution

Marks
3

$$\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos^2 3x \, dx$$

$$\cos 6x = 2 \cos^2 3x - 1$$

$$2 \cos^2 3x = \cos 6x + 1$$

$$\cos^2 3x = \frac{\cos 6x}{2} + \frac{1}{2}$$

$$\therefore \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos^2 3x \, dx = \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \left(\frac{\cos 6x}{2} + \frac{1}{2} \right) dx$$

$$= \frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (\cos 6x + 1) dx$$

$$= \frac{1}{2} \left[\frac{\sin 6x}{6} + x \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

$$= \frac{1}{2} \left[\left(\frac{\sin 6\left(\frac{\pi}{2}\right)}{6} + \frac{\pi}{2} \right) - \left(\frac{\sin 6\left(\frac{\pi}{3}\right)}{6} + \frac{\pi}{3} \right) \right]$$

$$= \frac{1}{2} \left[\left(0 + \frac{\pi}{2} \right) - \left(0 + \frac{\pi}{3} \right) \right]$$

$$= \frac{1}{2} \times \left(\frac{\pi}{2} - \frac{\pi}{3} \right)$$

$$= \frac{1}{2} \times \frac{\pi}{6}$$

$$= \frac{\pi}{12}$$

(e)

3

$$\frac{x^2 + x - 6}{x} \geq 2$$

$$x^2 \times \left(\frac{x^2 + x - 6}{x} \right) \geq 2 \times x^2$$

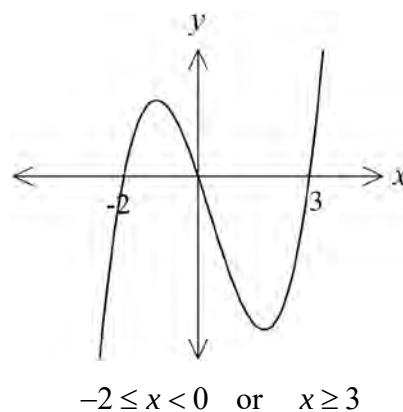
$$x(x^2 + x - 6) \geq 2x^2$$

$$x(x^2 + x - 6) - 2x^2 \geq 0$$

$$x(x^2 + x - 6 - 2x) \geq 0$$

$$x(x^2 - x - 6) \geq 0$$

$$x(x-3)(x+2) \geq 0$$



11.
(f)

Solution

Marks
2

$$y_1 = \sin x \quad \text{and} \quad y_2 = \cos x$$

$$\frac{dy}{dx_1} = \cos x \quad \text{and} \quad \frac{dy}{dx_2} = -\sin x$$

Curves intersect at $P\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$

$$\text{At } x = \frac{\pi}{4}$$

$$\frac{dy}{dx_1} = \cos\left(\frac{\pi}{4}\right) \quad \text{and} \quad \frac{dy}{dx_2} = -\sin\left(\frac{\pi}{4}\right)$$

$$m_1 = \frac{1}{\sqrt{2}} \quad \text{and} \quad m_2 = -\frac{1}{\sqrt{2}}$$

$$\tan \alpha = \left| \frac{\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}} \times -\frac{1}{\sqrt{2}}} \right|$$

$$= \left| \frac{\frac{2}{\sqrt{2}}}{1 - \frac{1}{2}} \right|$$

$$= \left| \frac{2}{\sqrt{2}} \div \frac{1}{2} \right|$$

$$= \left| \frac{2}{\sqrt{2}} \times \frac{2}{1} \right|$$

$$= \frac{4}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

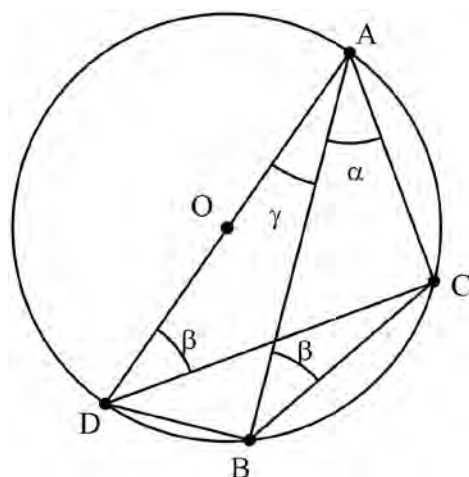
$$= \frac{4\sqrt{2}}{2}$$

$$= 2\sqrt{2}$$

11.
(g)

Solution

Marks
2



$$\angle ABC = \beta \text{ (given)}$$

$$\angle ABC = \angle ADC \left(\begin{array}{l} \text{angles subtended by the} \\ \text{same arc are equal} \end{array} \right)$$

$$\therefore \angle ADC = \beta$$

$$\angle ACD = 90^\circ \left(\begin{array}{l} \text{angle at the centre is twice the} \\ \text{angle at the circumference, diameter is } 180^\circ \end{array} \right)$$

$$\angle ADC + \angle DCA + \angle CAD = 180^\circ \text{ (angle sum of a triangle is } 180^\circ)$$

$$\angle DAB = \gamma \text{ and } \angle BAC = \alpha \text{ (given)}$$

$$\angle DAC = \alpha + \gamma \text{ (adjacent angles)}$$

$$\therefore \beta + 90^\circ + (\alpha + \gamma) = 180^\circ$$

$$\therefore \alpha + \beta + \gamma = 90^\circ$$

OR

$ABCD$ is a cyclic quadrilateral

$$\therefore \angle DAC + \angle DBC = 180^\circ \left(\begin{array}{l} \text{opposite angles in a} \\ \text{cyclic quadrilateral are supplementary} \end{array} \right)$$

$$\angle ABD = 90^\circ \text{ (angle in a semicircle)}$$

$$\angle DAB = \gamma, \angle BAC = \alpha, \angle ABC = \beta \text{ (given)}$$

$$\angle DAC = \angle DAB + \angle BAC \text{ (adjacent angles)}$$

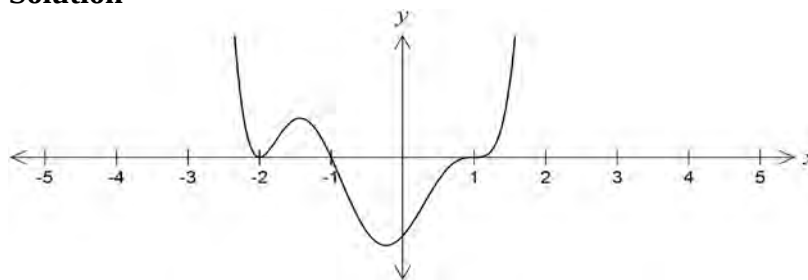
$$\therefore \gamma + \alpha + \beta + 90^\circ = 180^\circ$$

$$\therefore \alpha + \beta + \gamma = 90^\circ$$

12.
(a)

Solution

Marks
1



Any equation of the form

$$y = k(x+2)^2(x+1)(x-1)^3$$

where $k > 0$

(b)

Prove that,

3

$$1 + (1+2) + \dots + (1+2+\dots+2^{n-1}) = 2^{n+1} - n - 2$$

Prove true for $n = 1$

$$LHS = 1$$

$$RHS = 2^{1+1} - 1 - 2$$

$$= 4 - 1 - 2$$

$$= 1$$

$\therefore$ it is true for $n = 1$

Assume true for $n = k$

$$1 + (1+2) + \dots + (1+2+\dots+2^{k-1}) = 2^{k+1} - k - 2$$

Prove true for $n = k + 1$

i.e. Prove that

$$1 + (1+2) + \dots + (1+2+\dots+2^{k-1}) + (1+2+\dots+2^{k-1}+2^{k-1+1}) = 2^{(k+1)+1} - (k+1) - 2$$

$$1 + (1+2) + \dots + (1+2+\dots+2^{k-1}) + (1+2+\dots+2^{k-1}+2^k) = 2^{k+2} - (k+1) - 2$$

$$LHS = 1 + (1+2) + \dots + (1+2+\dots+2^{k-1}) + (1+2+\dots+2^{k-1}+2^k)$$

$$= 2^{k+1} - k - 2 + (1+2+\dots+2^{k-1}+2^k) \text{ by assumption}$$

Now, $1+2+\dots+2^{k-1}+2^k$ is a GP

$$a = 1, r = 2, n = k + 1$$

$$S_k = \frac{1 \times (2^{k+1} - 1)}{2 - 1}$$

$$= 2^{k+1} - 1$$

$$LHS = 2^{k+1} - k - 2 + 2^{k+1} - 1$$

$$= 2 \times 2^{k+1} - k - 3$$

$$= 2^1 \times 2^{k+1} - (k+1) - 2$$

$$= 2^{k+2} - (k+1) - 2$$

$$= RHS$$

$\therefore$ Proved true by mathematical induction

12.	Solution	Marks
(c)	(i) $x^3 + x^2 - 4x - k^2 = 0$ Let the roots of the equation be α, β and γ $\gamma = \alpha\beta$ $a = 1, b = 1, c = -4$ and $d = -k^2$ Product of the roots $\alpha\beta\gamma = -\frac{d}{a}$ $\gamma\gamma = -\frac{-k^2}{1}$ $\gamma^2 = k^2$ Since $k > 0$ and all $\alpha, \beta, \gamma > 0$ $\gamma = k$ $\therefore x = k$ is a root of the equation	2
(ii)	Sum of the roots 1 at a time $\alpha + \beta + k = -\frac{b}{a}$ $\alpha + \beta + k = -1$ $\alpha + \beta = -k - 1$ Sum of the roots 2 at a time $k + \alpha k + \beta k = \frac{c}{a}$ $k(1 + \alpha + \beta) = \frac{-4}{1}$ $k(1 + -k - 1) = -4$ $-k^2 = -4$ $\therefore k^2 = 4$ $\therefore k = 2 \text{ (since } k > 0)$ OR $k^3 + k^2 - 4k - k^2 = 0$ $k^3 - 4k = 0$ $k(k + 2)(k - 2) = 0$ $\therefore k = 2 \text{ only as } k > 0$	2

12.	Solution	Marks
(d)	$x^2 - \sqrt{x} - 2 = 0$ $\text{Let } f(x) = x^2 - \sqrt{x} - 2$ $= x^2 - x^{\frac{1}{2}} - 2$ $f'(x) = 2x - \frac{1}{2}x^{-\frac{1}{2}}$ $f'(2) = 2 \times 2 - \frac{1}{2\sqrt{2}}$ $f(2) = 2^2 - \sqrt{2} - 2$ $a_2 = a_1 - \frac{f(x)}{f'(x)}$ $= 2 - \frac{2 - \sqrt{2}}{4 - \frac{1}{2\sqrt{2}}}$ ≈ 1.84	2
(e)	(i) $x^2 = 8y$ $y = \frac{x^2}{8}$ $y' = \frac{2x}{8}$ $= \frac{x}{4}$ at $x = 4p$ $y' = \frac{4p}{4}$ $= p$ $y - 2p^2 = p(x - 4p)$ $y - 2p^2 = px - 4p^2$ $\therefore$ tangent at P is $y = px - 2p^2$	1

12.	Solution	Marks
(ii)	Tangent at Q is $y = qx - 2q^2 \dots\dots(2)$ $y = px - 2p^2 \dots\dots(1)$ $(1) - (2) \quad 0 = (p - q)x - 2(p^2 - q^2)$ $(p - q)x = 2(p - q)(p + q)$ $x = 2(p + q)$ sub into (1) $y = p \times 2(p + q) - 2p^2$ $y = 2p^2 + 2pq - 2p^2$ $y = 2pq$ $\therefore T$ is $(2(p + q), 2pq)$	1
(iii)	$P(4p, 2p^2)$ and $Q(4q, 2q^2)$ $m = \frac{2p^2 - 2q^2}{4p - 4q}$ $= \frac{2(p + q)(p - q)}{4(p - q)}$ $= \frac{p + q}{2}$ $y - 2p^2 = \frac{p + q}{2}(x - 4p)$ $2y - 4p^2 = (p + q)(x - 4p)$ $2y - 4p^2 = px - 4p^2 + qx - 4pq$ $2y = (p + q)x - 4pq$	1
(iv)	Since PQ passes through $A(0,1)$ $2 \times 1 = (p + q) \times 0 - 4pq$ $2 = -4pq$ $pq = -\frac{1}{2}$	2

13.	Solution	Marks
(a)	(i) $\triangle OWX \parallel \triangle OYZ$ (equiangular triangles are similar) $\therefore \frac{x}{x+d} = \frac{r}{R} \left(\begin{array}{l} \text{corresponding sides of similar triangles} \\ \text{are in proportion} \end{array} \right)$ $\frac{R}{r} = \frac{x+d}{x}$ $R = \frac{r(x+d)}{x}$ Area of the shadow is given by $A = \pi R^2$ $= \pi \left[\frac{r(x+d)}{x} \right]^2$ $= \frac{\pi r^2 (x+d)^2}{x^2}$	2
(ii)	$\frac{dx}{dt} = -4$ $A = \frac{\pi r^2 (x+d)^2}{x^2}$ $= \frac{\pi r^2 (x^2 + 2xd + d^2)}{x^2}$ $= \frac{\pi r^2 x^2 + 2\pi r^2 dx + \pi r^2 d^2}{x^2}$ $= \pi r^2 + 2\pi r^2 dx^{-1} + \pi r^2 d^2 x^{-2}$ $\frac{dA}{dx} = -2\pi r^2 dx^{-2} - 2\pi r^2 d^2 x^{-3}$ $\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt}$ $= \left(-\frac{2\pi r^2 d}{x^2} - \frac{2\pi r^2 d^2}{x^3} \right) \times -4$ When $x = d$ $\frac{dA}{dt} = \left(-\frac{2\pi r^2 d}{d^2} - \frac{2\pi r^2 d^2}{d^3} \right) \times -4$ $= \frac{8\pi r^2}{d} + \frac{8\pi r^2}{d}$ $= \frac{16\pi r^2}{d}$	2

13.
(b)

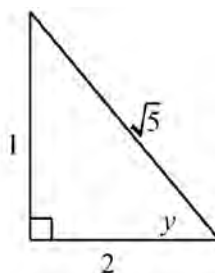
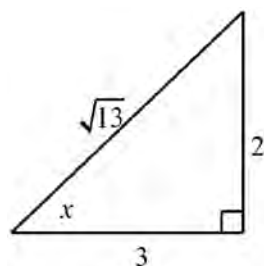
Solution

Marks
3

$$\tan^{-1} \frac{2}{3} + \cos^{-1} \frac{2}{\sqrt{5}} = \tan^{-1} \frac{7}{4}$$

$$\text{Let } x = \tan^{-1} \frac{2}{3} \text{ and } y = \cos^{-1} \frac{2}{\sqrt{5}}$$

$$\therefore \tan x = \frac{2}{3} \text{ and } \tan y = \frac{2}{\sqrt{5}}$$



$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$= \frac{\frac{2}{3} + \frac{1}{2}}{1 - \frac{2}{3} \times \frac{1}{2}}$$

$$= \frac{7}{4}$$

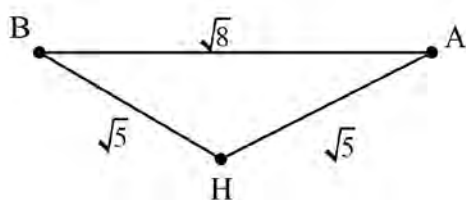
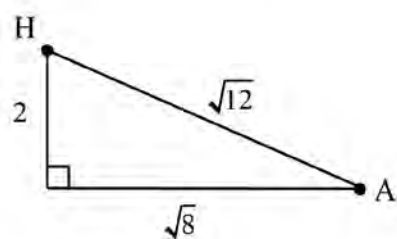
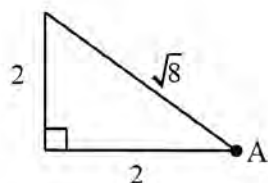
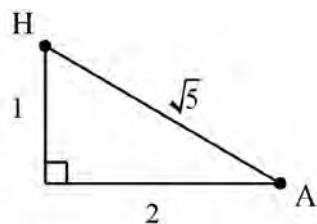
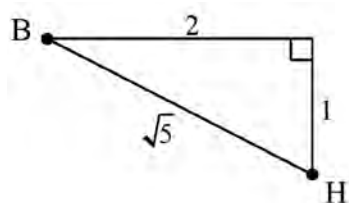
$$\therefore x + y = \tan^{-1} \frac{7}{4}$$

$$\therefore \tan^{-1} \frac{2}{3} + \cos^{-1} \frac{2}{\sqrt{5}} = \tan^{-1} \frac{7}{4}$$

13.
(c)

Solution

Marks
2



$$\begin{aligned}\cos(\angle AHB) &= \frac{(\sqrt{5})^2 + (\sqrt{5})^2 - (\sqrt{8})^2}{2 \times \sqrt{5} \times \sqrt{5}} \\ &= \frac{5 + 5 - 8}{2 \times 5} \\ &= -\frac{1}{5} \\ \therefore \angle AHB &= 101.5^\circ\end{aligned}$$

13.

(d) (a)

Solution**Marks**

2

$$\frac{\sin 4t}{\sqrt{3}} - \cos 4t = R \sin(4t - \alpha)$$

$$R \sin(4t - \alpha) = R(\sin 4t \cos \alpha - \cos 4t \sin \alpha)$$

$$R \sin 4t \cos \alpha - R \cos 4t \sin \alpha = \frac{\sin 4t}{\sqrt{3}} - \cos 4t$$

Equating co-efficients

$$R \sin \alpha = 1 \dots\dots(1)$$

$$R \cos \alpha = \frac{1}{\sqrt{3}} \dots\dots(2)$$

$$(1) \div (2) \dots \tan \alpha = \frac{1}{\frac{1}{\sqrt{3}}} \\ = \sqrt{3}$$

$$\alpha = \frac{\pi}{3}$$

$$R^2 \sin^2 \alpha = 1^2 \dots\dots(1)^2$$

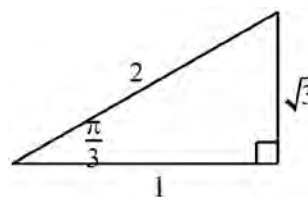
$$R^2 \cos^2 \alpha = \left(\frac{1}{\sqrt{3}}\right)^2 \dots\dots(2)^2$$

$$(1)^2 + (2)^2 \dots R^2 \sin^2 \alpha + R^2 \cos^2 \alpha = 1 + \frac{1}{3}$$

$$R^2 \times 1 = \frac{4}{3}$$

$$R = \frac{2}{\sqrt{3}}$$

$$\therefore \frac{\sin 4t}{\sqrt{3}} - \cos 4t = \frac{2}{\sqrt{3}} \sin\left(4t - \frac{\pi}{3}\right)$$



(b)

$$x = 4 + \frac{\sin 4t}{\sqrt{3}} - \cos 4t$$

$$\dot{x} = \frac{4 \cos 4t}{\sqrt{3}} + 4 \sin 4t$$

$$\ddot{x} = \frac{-16 \sin 4t}{\sqrt{3}} + 16 \cos 4t$$

$$= -16 \left(\frac{\sin 4t}{\sqrt{3}} - \cos 4t \right)$$

$$= -16 \left(4 + \frac{\sin 4t}{\sqrt{3}} - \cos 4t - 4 \right)$$

$$= -16(x - 4)$$

2

13.

Solution

Marks

(c)

$$x = \frac{2}{\sqrt{3}} \sin\left(4t - \frac{\pi}{3}\right)$$

$$\dot{x} = 4 \times \frac{2}{\sqrt{3}} \cos\left(4t - \frac{\pi}{3}\right)$$

Maximum speed occurs when $\cos\left(4t - \frac{\pi}{3}\right) = 1$

$\therefore$ maximum speed is $\frac{8}{\sqrt{3}}$ or $\frac{8\sqrt{3}}{3}$

2

14.

Solution

Marks

(a)

2

$$(1+x)^{n+4} = (1+x)^n (1+x)^4$$

$$LHS = (1+x)^{n+4}$$

$$= \binom{n+4}{0} + \binom{n+4}{1}x + \binom{n+4}{2}x^2 + \dots + \binom{n+4}{r}x^r + \dots + \binom{n+4}{n+4}x^{n+4}$$

$$RHS = (1+x)^n (1+x)^4$$

$$= \left[\binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{r}x^r + \dots + \binom{n}{n}x^n \right]$$

$$\left[\binom{4}{0} + \binom{4}{1}x + \binom{4}{2}x^2 + \binom{4}{3}x^3 + \binom{4}{4}x^4 \right]$$

Equating co-efficients of x^r

$$\binom{n+4}{r} = \binom{4}{0}\binom{n}{r} + \binom{4}{1}\binom{n}{r-1} + \binom{4}{2}\binom{n}{r-2} + \binom{4}{3}\binom{n}{r-3} + \binom{4}{4}\binom{n}{r-4}$$

$$= \binom{n}{r} + 4\binom{n}{r-1} + 6\binom{n}{r-2} + 4\binom{n}{r-3} + \binom{n}{r-4}$$

(b)

2

$$\int (\tan^3 x \sec^2 x) dx$$

$$u = \tan x$$

$$\frac{du}{dx} = \sec^2 x$$

$$du = \sec^2 x dx$$

$$\therefore \int (\tan^3 x \sec^2 x) dx = \int u^3 du$$

$$= \left[\frac{u^4}{4} \right] + C$$

$$= \frac{\tan^4 x}{4} + C$$

14.	Solution	Marks
(c)	(i) Range is all real $y > 0$	1
	(ii) $f(x) = x + \sqrt{x^2 + 1}$	2
	$f(x) = x + (x^2 + 1)^{\frac{1}{2}}$	
	$f'(x) = 1 + \frac{1}{2}(x^2 + 1)^{-\frac{1}{2}} \times 2x$	
	$= 1 + \frac{x}{\sqrt{x^2 + 1}}$	
	$= \frac{\sqrt{x^2 + 1}}{\sqrt{x^2 + 1}} + \frac{x}{\sqrt{x^2 + 1}}$	
	$= \frac{x + \sqrt{x^2 + 1}}{\sqrt{x^2 + 1}}$	
	$= \frac{f(x)}{\sqrt{x^2 + 1}}$	
(iii)	Range is all real y	1
(iv)	$f : y = x + \sqrt{x^2 + 1}$	2
	$f^{-1} : x = y + \sqrt{y^2 + 1}$	
	$x - y = \sqrt{y^2 + 1}$	
	$(x - y)^2 = y^2 + 1$	
	$x^2 - 2xy + y^2 = y^2 + 1$	
	$x^2 - 2xy = 1$	
	$2xy = x^2 - 1$	
	$y = \frac{x^2 - 1}{2x}$	
	$= \frac{1}{2} \left(x - \frac{1}{x} \right)$	

14.		Solution	Marks
(d)	(i)	Maximum height occurs when $\dot{y} = 0$	2
		$y = Vt \sin \theta - \frac{1}{2}gt^2$	
		$\dot{y} = V \sin \theta - gt$	
		$V \sin \theta - gt = 0$	
		$gt = V \sin \theta$	
		$t = \frac{V \sin \theta}{g}$	
		at $t = \frac{V \sin \theta}{g}$	
		$y = V \left(\frac{V \sin \theta}{g} \right) \sin \theta - \frac{1}{2}g \left(\frac{V \sin \theta}{g} \right)^2$	
		$= \frac{V^2 \sin^2 \theta}{g} - \frac{V^2 \sin^2 \theta}{2g}$	
		$= \frac{2V^2 \sin^2 \theta}{2g} - \frac{V^2 \sin^2 \theta}{2g}$	
		$= \frac{V^2 \sin^2 \theta}{2g}$	

(ii) Projectile reaches its target when $y = -h$

2

$$Vt \sin \theta - \frac{1}{2}gt^2 = -h$$

$$Vt \sin \theta - \frac{1}{2}gt^2 = -\frac{V^2 \sin^2 \theta}{2g}$$

$$\frac{2Vt \sin \theta g}{g} - \frac{g^2 t^2}{g} + \frac{V^2 \sin^2 \theta}{g} = 0$$

$$g^2 t^2 - 2V \sin \theta g t - V^2 \sin^2 \theta = 0$$

$$\begin{aligned}\therefore t &= \frac{-(-2V \sin \theta g) \pm \sqrt{(-2V \sin \theta g)^2 - 4 \times g^2 \times -V^2 \sin^2 \theta}}{2g^2} \\ &= \frac{2V \sin \theta g \pm \sqrt{4V^2 \sin^2 \theta g^2 + 4g^2 V^2 \sin^2 \theta}}{2g^2}\end{aligned}$$

Since t cannot be negative

$$\begin{aligned}t &= \frac{2V \sin \theta g + \sqrt{8V^2 \sin^2 \theta g^2}}{2g^2} \\ &= \frac{2V \sin \theta g + 2\sqrt{2}(Vg \sin \theta)}{2g^2} \\ &= \frac{V \sin \theta + \sqrt{2}(V \sin \theta)}{g} \\ &= \frac{V \sin \theta (1 + \sqrt{2})}{g} \text{ seconds}\end{aligned}$$

(iii)

When the projectile reaches the target, $t = \frac{V \sin \theta (1 + \sqrt{2})}{g}$ seconds

1

$$\begin{aligned}x &= V \frac{V \sin \theta (1 + \sqrt{2})}{g} \cos \theta \\ &= \frac{V^2 \sin \theta \cos \theta (1 + \sqrt{2})}{g} \times \frac{2}{2} \\ &= \frac{V^2 2 \sin \theta \cos \theta (1 + \sqrt{2})}{2g} \\ &= \frac{V^2 (1 + \sqrt{2}) \sin 2\theta}{g}\end{aligned}$$